Loan Default Prediction

An interpretable credit-scoring model for home equity loan approvals

Capstone Project | Practical Data Science | MIT Professional Education - Applied AI and Data Science Program

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Key takeaways

1 in 5

home equity loans in the data ended in default. Accuracy alone is a misleading yardstick.

62%

default rate when the applicant's debt-to-income ratio is missing, versus 8.6% when it is recorded. Unverified income is the strongest warning sign.

93%

of future defaulters are caught by the proposed Random Forest model on unseen applicants, while 2 of every 3 applicants it flags really would have defaulted.

~$9M

indicative annual benefit on 6,000 applications, from roughly $10.8M of default-related cost down to about $1.8M.

The problem: approving home equity loans without losing the profit to defaults

- Interest on home loans is a major share of retail bank profit; bad loans eat it up.
- Today each application is judged manually: slow, inconsistent across underwriters, open to error and bias.
- The consumer credit department wants an empirically derived credit-scoring model built on its own recent loans.
- Constraint: under the Equal Credit Opportunity Act every rejection needs a specific, defensible reason, so the model must be interpretable.

Objective: predict which applicants are likely to default and tell the bank which attributes to weigh when approving a loan.

The data: Home Equity dataset (HMEQ)

5,960
recent loans

12
attributes known at application

19.9%
defaulted or seriously delinquent

44%
of applications have a blank field

Approach: from raw applications to a decision the bank can defend

1

Explore

Univariate, bivariate and correlation analysis; default rate by band; pattern of missing values

2

Prepare

Hold out a stratified 30% test set first; flag every missing field; impute median/mode learned on training rows; cap outliers for the linear model only; one-hot encode

3

Model

Logistic Regression (linear benchmark), Decision Tree, Bagging, Random Forest; class weights; cross-validated grid search

4

Measure

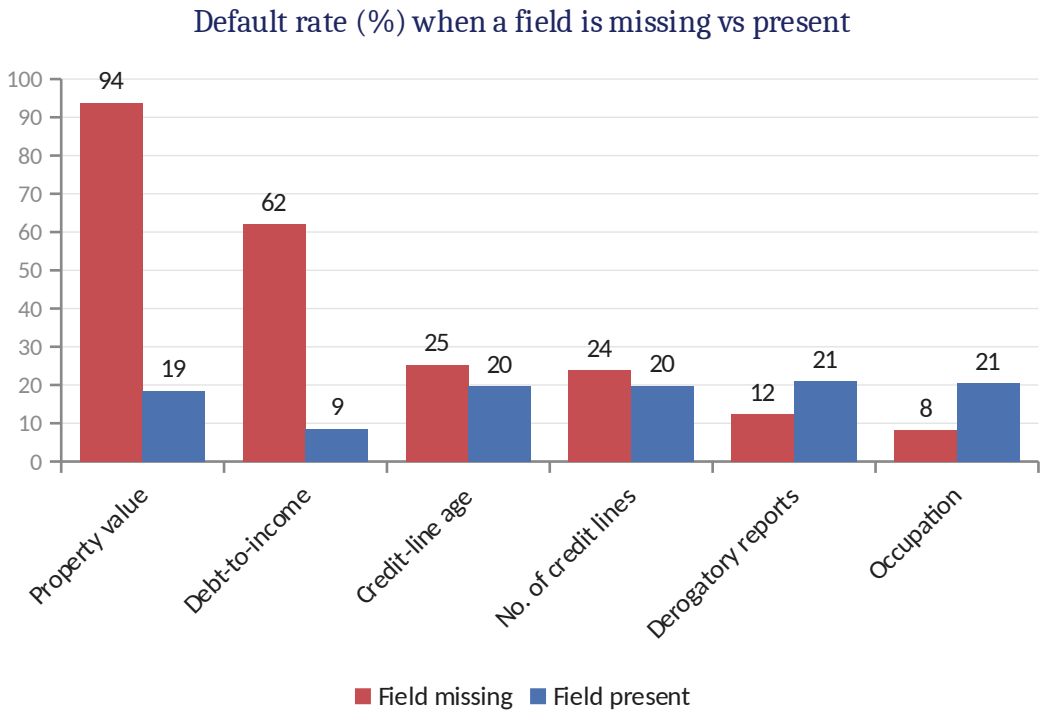
Recall on defaulters as the primary metric; precision, F1, ROC-AUC as guard-rails; final model and threshold chosen on out-of-fold expected cost

5

Recommend

Model specification, business rules, implementation plan, benefits, risks

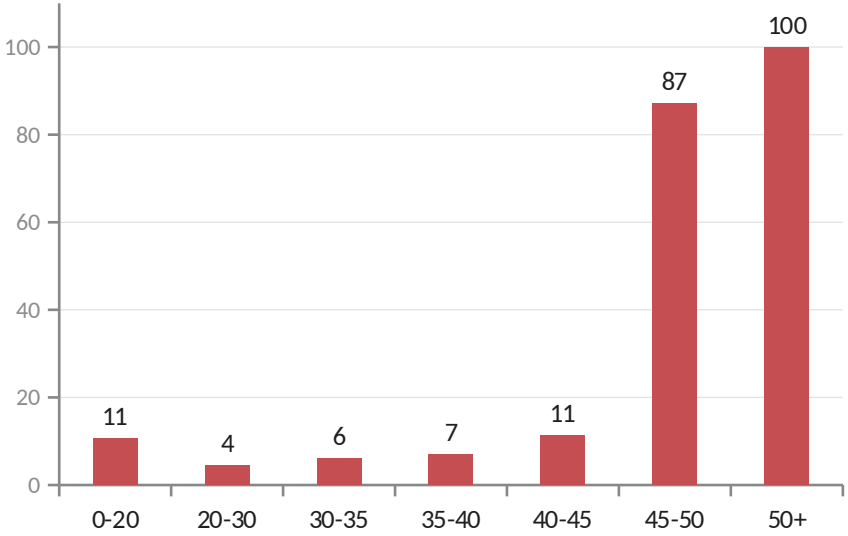
Insight 1: information the applicant did not provide is the strongest warning sign



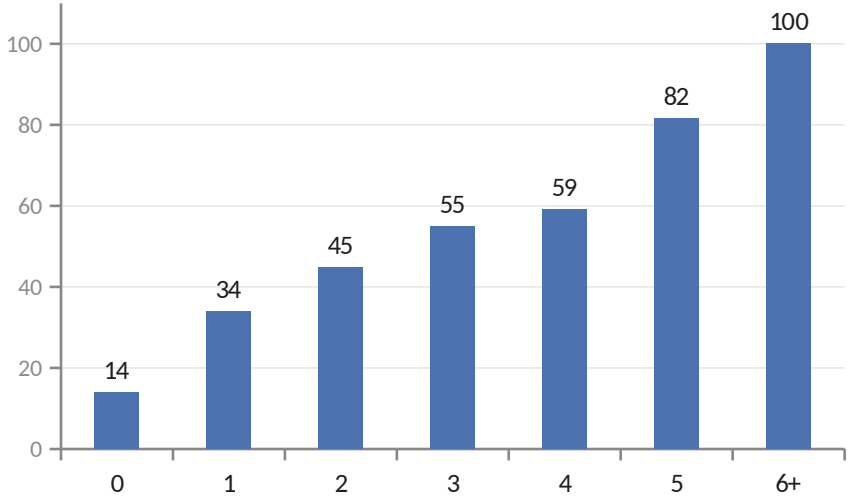
- No recorded debt-to-income ratio: 62% default (1,267 applicants, 21% of the data).
- No property valuation: 94% default (112 applicants).
- Overall default rate is 20%, so these gaps multiply risk three to five times.
- Interpretation: income or collateral that could not be verified, or that the applicant chose not to supply.
- Action: every imputed field gets a 'was missing' flag in the model, and verification should become mandatory in the process.

Insight 2: debt burden has a cliff at 45%, and past delinquency repeats

Default rate (%) by debt-to-income ratio band (%)



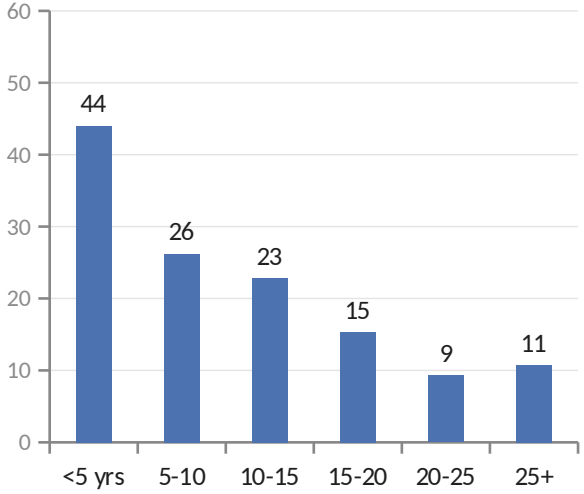
Default rate (%) by number of delinquent credit lines



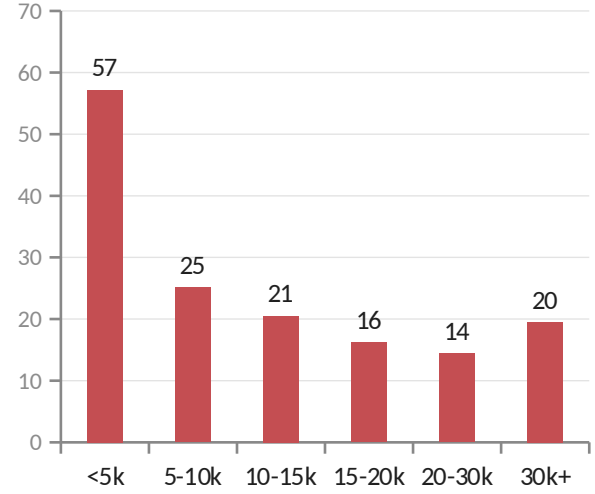
Between a 20% and a 40% ratio only 4-7% default; above 45% almost everyone does. Each delinquent line or derogatory report roughly doubles the risk (0 reports: 17%, 1: 39%, 2: 51%, 3+: about 75%). Both are plain-language rules an applicant can be given.

Insight 3: long credit histories protect, small loans do not, occupation matters less

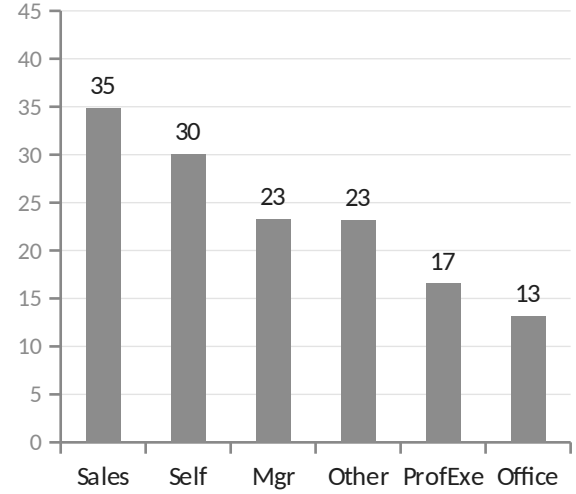
By age of oldest credit line



By loan amount ($)



By occupation



Default rate (%) by group. Applicants with under five years of credit history default four times as often as those with twenty-plus years. Loans under $5,000 default 57% of the time. Occupation shifts risk moderately; on its own it adds little at the usual cut-off, but removing it from the final model raises the expected cost of credit errors by about 10%, so it stays in, subject to the fair-lending review.

Measure of success: the two mistakes do not cost the same

Missed defaulter

The model says 'will repay', the customer defaults.

Bank loses most of the principal plus collection cost: about $9,000 on a typical $16,900 defaulted loan.

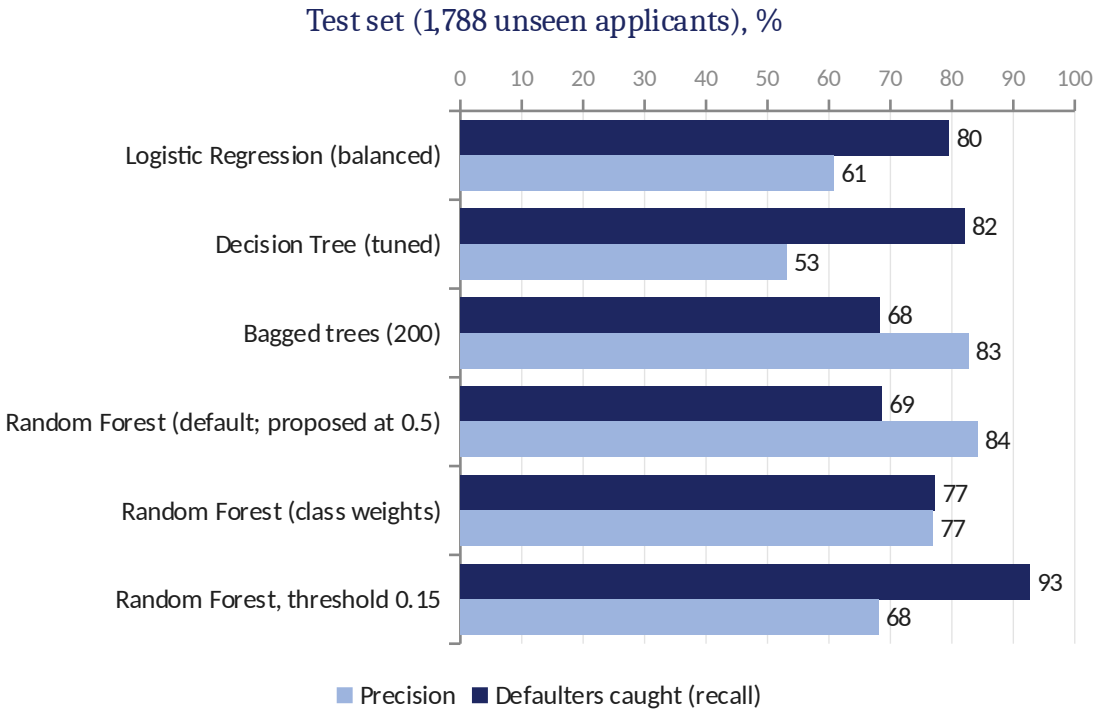
Wrongly declined

The model says 'will default', the customer would have repaid.

Bank forgoes the interest margin on one good customer: about $2,000.

- A missed defaulter costs roughly 4.5 times a wrongly declined customer, so the primary metric is Recall: the share of future defaulters the model catches.
- Guard-rails: Precision (how many of the flagged applicants really default) and F1, so the model does not simply decline everyone. ROC-AUC measures how well applicants are ranked by risk, which matters when the threshold moves.
- Individual models are tuned by cross-validated grid search on recall; the final model and its threshold are chosen on out-of-fold expected cost, training data only; the held-out 30% is scored once.

Model comparison: the Random Forest gives the best balance



- Linear vs non-linear: logistic regression reaches 80% recall but only 61% precision. The data is made of thresholds, which trees capture and a straight line cannot.
- Single tree vs ensembles: the tuned tree catches 82% but declines almost one good customer per defaulter caught (53% precision). Bagging 200 trees lifts precision to 83% and AUC to 0.95 at 68% recall.
- The class-weighted forest: 77% recall at 77% precision, ROC-AUC 0.96, among the best at ranking applicants by risk.
- Final choice on the expected cost of mistakes (training data only), by a rule fixed beforehand: lowest average cost, simplest if within noise. Three forests tied; the plain forest at 0.15 catches 93% at 68% precision on test.

Proposed solution: a Random Forest scorer explained by a surrogate decision tree

Model specification

- Scoring engine: Random Forest, 200 fully grown trees, no class weights: the decision threshold handles the imbalance.
- Inputs: all 27 (10 financial attributes, 11 missing-value flags, occupation and loan purpose). Fallback without occupation: 21 inputs, about 10% more credit-error cost.
- Decision threshold 0.15, chosen together with the configuration by minimising expected cost on repeated out-of-fold predictions of the training rows (stable for cost ratios 3:1 to 6:1 and across train/test splits); approve / refer / decline bands (about 69% / 7% / 24% of applicants) keep an underwriter in the loop.
- Explanation layer: a depth-4 surrogate tree fitted to the forest's out-of-fold decisions reproduces 93% of them; its rules (ratio missing; ratio above 44%; a delinquent line together with a derogatory report) supply the ECOA adverse-action reasons.
- Test set held out before any pre-processing; scored once, after all choices were made on training data.

93%
of defaulters caught

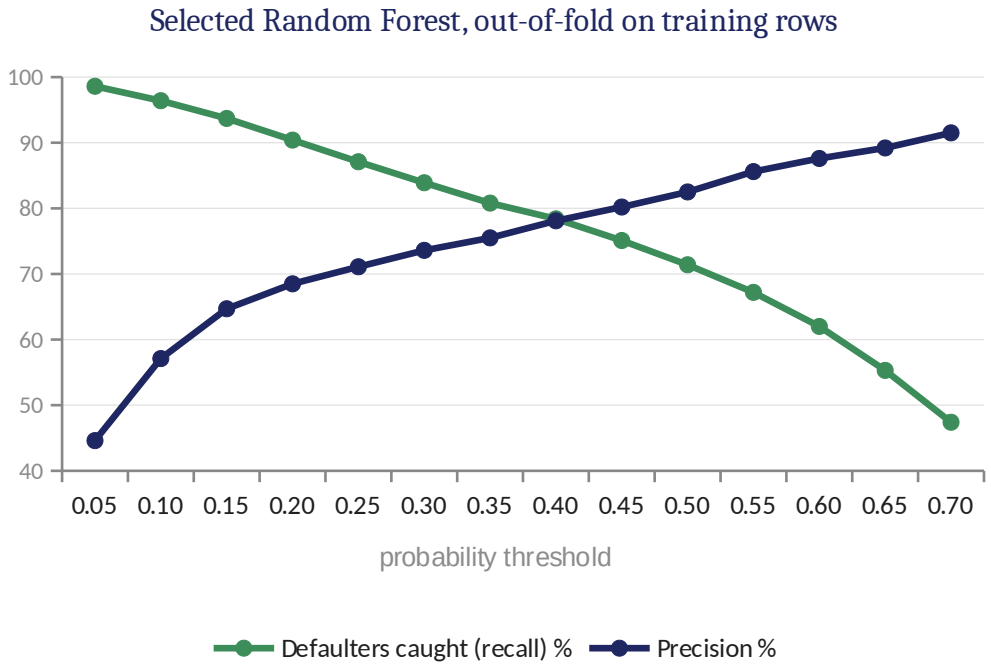
68%
of flagged applicants truly default

0.96
ROC-AUC: risk ranking quality

90%
overall accuracy

Per 1,000 applicants: 185 of 200 defaulters flagged in advance, 15 missed, 87 good customers referred or declined; 27% of applicants flagged, 73% approved.

Why a 0.15 threshold: the decision cut-off is a business dial, not a constant



At 0.50 (default)

- 63 missed defaulters
- 26 good customers declined
- $615k indicative cost

At 0.15 (proposed)

- 15 missed defaulters
- 87 good customers declined
- $304k indicative cost

Curves: out-of-fold predictions on the training rows, which is where the threshold was chosen. Cards: test set, per 1,000 applicants; cost assumes $9,000 per missed defaulter and $2,000 per wrongly declined customer. The optimum stays at 0.15 for cost ratios from 3:1 to 6:1, on a finer grid and on other train/test splits; it sits just below the textbook break-even (0.18) because the scores understate risk.

Business impact: an indicative $9 million a year

Per year, 6,000 applications	Current process	With the proposed model
Loans that default	about 1,200 (20%)	about 90 slip through
Default-related losses	about $10.8M	about $0.8M
Good customers declined or referred	none	about 520, margin forgone about $1.0M
Total cost of credit errors	**about $10.8M**	**about $1.8M**

~$9M

indicative gross annual benefit; about $8.9M after running costs, $8.5M in year one

- Assumptions: 6,000 applications a year, defaulted loans average $16,900 with about half recovered ($9,000 per missed defaulter), forgone margin about $2,000 per declined good customer, production performance equal to test performance; measured as an extra screen on the loans the bank approves today, before mandatory income verification.
- Costs: roughly $300-400k one-off (pipeline, integration, validation, fairness review, training) and about $100k a year to run and monitor. Pays back within the first year even if only a quarter of the benefit is realised.
- Non-financial: faster decisions, consistent treatment, auditable reasons for every decline, verified income and valuation on every application.

Recommendations and implementation roadmap

1 Months 0-2: prepare

- Make income and valuation verification mandatory fields
- Build the scoring pipeline and reason codes
- Confirm the threshold with real loss and margin figures
- Fairness and model-risk review

2 Months 2-4: shadow and pilot

- Score every application alongside the manual decision
- Re-train once verified ratios flow in; tune the approve / refer / decline bands
- Train underwriters on reason codes and referrals

3 Month 4 onwards: scale

- Switch on automated approve / decline for clear cases
- Monthly monitoring of recall, precision, approval rate, missing fields
- Yearly re-training; threshold review with finance

Policy rules to adopt now, independent of the model

- Decline or escalate above a 45% debt-to-income ratio; refer to an underwriter with two or more delinquent credit lines or two or more major derogatory reports (45% and 51%+ observed default). Hard rules take precedence over the score.
- Treat an application with no verifiable income or property valuation as high risk, never as neutral.

Risks, challenges and further analysis

Risks and mitigations

- Selection bias: the data covers only loans the old process approved. Mitigate with a shadow period on all applications and reject inference.
- The 'ratio missing' signal will change once verification is mandatory: re-train on the verified population before go-live. On applicants with a recorded ratio the model already catches 83% at 80% precision.
- Economic drift: monitor monthly, re-train yearly, review the threshold.
- Fair lending: the 0.15 cut-off wrongly flags about 11% of creditworthy applicants (3% at 0.5), so disparate-impact testing before launch, annually and whenever the cut-off moves; a 21-input fallback without occupation is ready (about 10% more credit-error cost).
- Explainability and automation bias: tree-based reason codes, a referral band, documented overrides.

Further analysis

- Reject inference to cover applicants the old process declined.
- Engineered features: loan-to-value and payment-to-income ratios.
- Gradient boosting and probability calibration, so a score of 0.15 means exactly a 15% chance of default.
- Time-based validation once application dates are available.
- Risk-based pricing for the referral band; early-warning scoring on existing loans.
- Per-applicant SHAP-style reason codes in applicant-friendly language.

Thank you

Questions and discussion

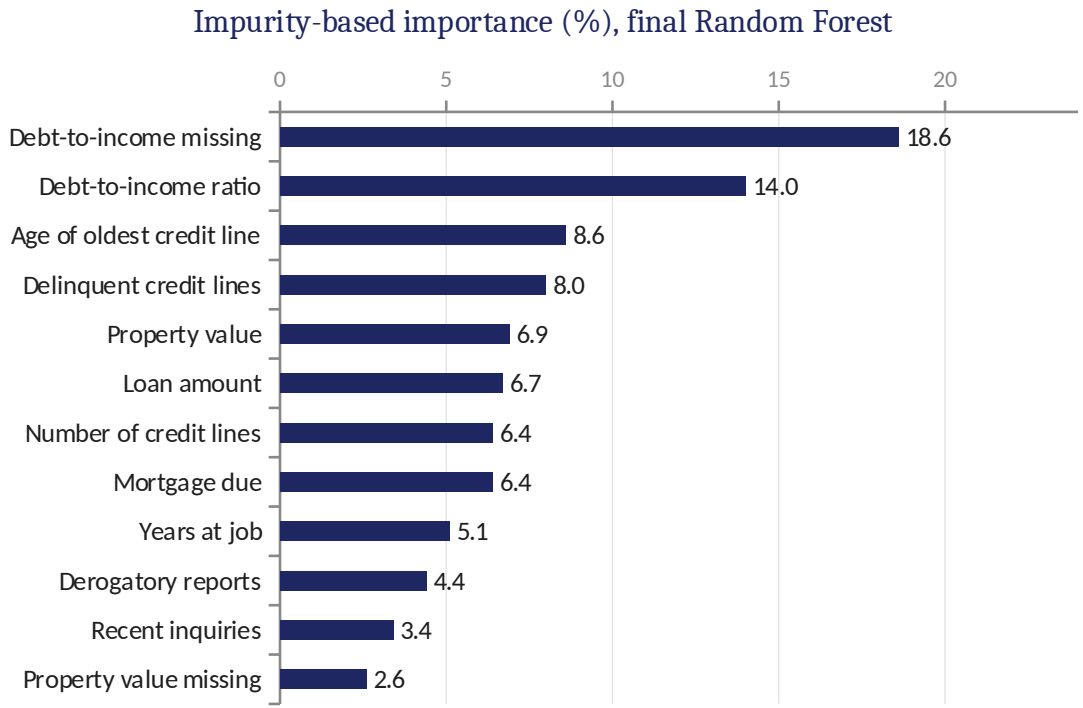
Catch 93% of defaulters, explain every decision, and save an indicative $9M a year.

Appendix A: all models on the held-out test set

Model	Recall	Precision	F1	Accuracy	ROC-AUC	Train recall
Logistic Regression (plain)	0.566	0.774	0.654	0.880	0.907	0.635
Logistic Regression (balanced, C tuned)	0.796	0.608	0.689	0.857	0.906	0.821
Decision Tree (default)	0.686	0.706	0.696	0.880	0.807	1.000
Decision Tree (class weights)	0.616	0.719	0.664	0.875	0.778	1.000
Decision Tree (tuned: balanced, min 20 per leaf)	0.821	0.531	0.645	0.819	0.877	0.921
Bagging (200 fully grown trees)	0.683	0.827	0.748	0.908	0.945	1.000
Bagging (class-weighted trees)	0.597	0.810	0.687	0.891	0.946	1.000
Random Forest (default, 200 trees)	0.686	0.842	0.756	0.912	0.963	1.000
Random Forest (class weights 0.2/0.8)	0.773	0.769	0.771	0.908	0.959	1.000
Random Forest (tuned for recall)	0.790	0.656	0.717	0.875	0.932	0.918
Random Forest (tuned, 12 features)	0.784	0.660	0.717	0.876	0.929	0.916
Proposed: RF unweighted, 27 inputs, threshold 0.15	**0.927**	**0.681**	**0.785**	**0.899**	**0.963**	-

Test set of 1,788 applicants at the 0.5 threshold except the proposed model. Train recall of 1.000 marks un-pruned models that memorised the training data. The proposed model (the default forest with its threshold moved) and its threshold were selected on repeated out-of-fold expected cost (training rows only) under a rule fixed beforehand: lowest mean cost, simplest set-up among those within one standard error (about $10,500). Mean out-of-fold ROC-AUC 0.961 against 0.963 on test.

Appendix B: what the Random Forest relies on



- The debt-to-income ratio and its missing flag carry a third of the final forest's decision power; with delinquent lines and credit-history length, about half.
- Permutation importance (drop in ranking quality when a column is shuffled) ranks the same four at the top, delinquent credit lines first.
- Loan, property and mortgage amounts matter in combination, little on their own.
- Occupation, loan purpose and minor flags matter little individually, yet dropping them raised out-of-fold expected cost by about a fifth, and dropping occupation alone by about 10%, so they stay subject to the fair-lending review.
- Logistic-regression coefficients, tree rules and the surrogate tree all point to the same drivers.

Appendix C: the explanation layer, rules of the surrogate tree that follows the forest

1	**Is the debt-to-income ratio missing?**	Yes: flag, whatever the credit history (62% of these applicants defaulted). Verify income before approving.
2	**If recorded, is the ratio above about 44%?**	Yes: flag (87-100% of applicants above 45% defaulted).
3	**Ratio at or below 44%: a delinquent credit line together with a major derogatory report?**	Yes: flag. Otherwise clear; applicants with no delinquent line default only about 4-5% of the time.
4	**How faithful are these rules?**	They reproduce 93% of the forest's decisions on unseen applicants (73% for never flagging; kappa 0.82); the forest's own top risk drivers explain the remaining 7%.
5	**What the rules do not use**	Loan size, credit-history length, occupation and loan purpose do not change the outcome in the surrogate: they matter only in combination inside the forest, which supplies the per-applicant reason codes.

Surrogate: depth-4 tree, minimum 20 applicants per leaf, fitted to the forest's out-of-fold flag/clear decisions at threshold 0.15 on the training rows; it agrees with the forest on 93% of test decisions (73% for a rule that never flags). The stand-alone tuned tree (test recall 82%, precision 53%) found the same drivers.

Appendix D: data treatment and pre-processing

Step	What was done	Why
Duplicates	None found; no rows dropped	5,960 loans retained, including 2 with only the loan amount recorded
Missing values	A 0/1 'was missing' flag per column, then median (numeric) or mode (categorical) imputation learned on the training rows	Missingness of debt-to-income and property value is the strongest default signal; dropping rows would lose 44% of the data
Outliers	IQR capping for the linear model only; DEROG and DELINQ capped at the 99th percentile; tree models use raw values	Linear models are pulled by extremes (debt-to-income 203%, credit history 97 years); trees split on rank order and keep the 'above 50% always defaults' signal
Encoding and scaling	One-hot encoding of REASON and JOB (drop first); standard scaling of the continuous inputs for the linear model only, flags left as 0/1	Comparable coefficients, odds ratios for flags read as the effect of the flag
Split	70% train / 30% test, stratified on the target, fixed seed, held out before any statistic was fitted; identical applicants in both data versions	No leakage from test rows into fences, medians or scaling; results comparable across models
Imbalance	Class weights (balanced or 0.2/0.8) inside the individual models, then the decision threshold for the final model; no resampling	Keeps every real applicant; class weights lifted recall by 23 points for logistic regression and 9 for the forest at the 0.5 cut-off, but made no measurable difference once the threshold was chosen on cost, so the final forest is unweighted
Model inputs	27 columns: the 10 attributes, 11 missing flags and 6 one-hot indicators (REASON HomeImp; JOB Office, Other, ProfExe, Sales, Self)	Every input to the final model is named in the notebook's final-pipeline cell, with the imputation values and model settings